# Daniel Engelberg

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Herzliya, Israel | Authorized to work in the US | Relocating to New York

## SUMMARY

Applied AI engineer (B.Sc., Technion) who builds agents for work where mistakes cost money: a permission-aware expense agent in TypeScript with an auditable run log, durable approval chains and decision evals (zero false approvals across 3 Claude models), plus a production WhatsApp AI platform in Python with 350+ tests.

## SKILLS

**AI:** Claude API (tool use, structured output, vision), agent loops, evals, prompt-injection defenses, OpenAI, OpenRouter, Claude Code (daily)
**Programming:** TypeScript, Python, JavaScript, Node.js, SQL
**Full stack:** Next.js, React, PostgreSQL (Drizzle, Alembic), Flask, REST APIs, webhooks, job queues, Vitest, pytest
**Cloud:** Vercel, Neon, Render, Google Cloud Run, Docker, Sentry, GitHub auto-deploy
**Languages:** English (native), Hebrew (fluent)

## EDUCATION

**Technion – Israel Institute of Technology, Haifa, Israel** Mar 2026
*B.Sc. in Mechanical Engineering, Robotics specialization*
**Relevant coursework:** Python, Robotics (Kinematics, Path Planning), Control Theory, Signal Processing, CAD, Mechanical Design, Manufacturing Processes, Fluid Mechanics, Solid Mechanics

## PROJECTS

**Expense Agent (permission-aware AI expense review)** | *TypeScript, Next.js, PostgreSQL, Claude API, Zod* Oct 2026

- Built an agent that reviews expense receipts: Claude vision extracts each receipt into a Zod-validated schema, a policy engine in code decides (limits, tip cap, alcohol, currency, duplicates, approval chains), and an LLM reviewer can only add human approval, never approve or change amounts
- Scoped every agent tool to the permissions of the person it acts for, so a lookup of a colleague's salary is denied and logged; nobody approves their own expense; every model call, tool call and approval attempt lands in an auditable run log with tokens, cost and latency
- Built decision evals over 42 synthetic receipts, graded on outcome and approvers with false approvals counted first: Claude Opus 5.5 and Sonnet 5.5 42/42, Haiku 5.5 40/42, zero false approvals, 3/3 prompt injections held; an eval-driven prompt fix took Sonnet from 37 to 42
- Ran each claim as a durable state machine on a Postgres job queue (SKIP LOCKED, retries with backoff, crash recovery from the last committed stage) with pause-and-resume approval chains across manager, IT and Finance
- Deployed on Vercel and Neon with demo guardrails (daily and per-visitor claim caps, nightly reset) and 108 automated tests; live at expense-agent-seven.vercel.app

**Handy AI** | *Python, Flask, PostgreSQL, Claude / LLM APIs, WhatsApp Cloud API* Oct 2026 – Present

- Built a WhatsApp AI maintenance coordinator for property managers: tenants report issues by text, photo or voice note, and the system runs intake, collects contractor quotes, schedules visits and requests payment
- Integrated Claude (Anthropic API) and OpenRouter with Pydantic-validated structured output, versioned prompts and per-call cost tracking; model output never changes state directly
- Engineered a Postgres-backed job queue (retries, dead-letter, per-conversation ordering, deduplication) and a 13-state ticket lifecycle with a full audit trail
- Wrote 400+ automated tests across 22,000+ lines of Python; deployed the web service and background worker on Render

## EXPERIENCE

**FlowTransit, Tel Aviv, Israel** Jul 2026 – Sep 2026
*Intern*

- Improved GTFS and OpenStreetMap preprocessing so complete intracity bus lines are kept whenever any segment crosses a selected area, using map matching and network edge processing

**Formula Student Technion, Haifa, Israel** Oct 2023 – Apr 2024
*High Voltage Engineer*

- Co-designed a new high-voltage accumulator for the race car from scratch in SolidWorks, managing files in Kenesto and tasks in monday.com

**TAMID Group (student consulting), Technion chapter** | *Co-Founder and Co-President* Jul 2023 – Jul 2024